

(TR-104) PYTHON WITH AI/ML

Training Day 36 Report

Date: 2 March 2026

Topic Covered: RAG-Based AI Assistant Project Planning and Initial Development

The thirty-sixth day of training focused on starting a Retrieval-Augmented Generation (RAG) based AI Assistant project. A project discussion and planning meeting was conducted to understand the project requirements, workflow, and implementation strategy for building an AI assistant capable of answering queries from company documents.

- Attended a project planning and discussion meeting for the RAG-based AI Assistant system.
- Understood the workflow and requirements of a document-based AI assistant project.
- Discussed how Retrieval-Augmented Generation (RAG) combines retrieval systems with language models for intelligent question answering.
- Planned the implementation flow for document loading, preprocessing, chunking, embeddings, retrieval, and response generation.
- Created a clean folder structure for organizing project modules and resources.
- Created sample company data in HTML, PDF, and FAQ formats for testing the assistant.
- Loaded and extracted text content from HTML, PDF, and FAQ documents.
- Split large document text into smaller chunks for efficient semantic retrieval.
- Converted document chunks into vector embeddings for similarity-based search.
- Explored the role of vector databases in storing and retrieving embeddings efficiently.
- Strengthened understanding of RAG pipelines, semantic search, and AI assistant architecture used in modern NLP systems.

- **GitHub Link:**

<https://github.com/JaspinderKaurWalia26/project>